

REDAC

User Manual*

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1 Introduction

This user manual introduces REDAC, a web application that features a graphical user interface for the analysis of bulk RNA-Seq count data (non-negative integers). The application is developed in the R programming language (<http://cran.r-project.org/>), an open-source, object-oriented environment for statistical computing and graphics. REDAC integrates several RNA-Seq analysis tools available through Bioconductor (www.bioconductor.org).

Availability: REDAC is an open source software released under the GPL3 license.

Available for local use at: <https://github.com/franruss/REDAC>

Available for online use at: <https://frusso.shinyapps.io/REDAC>

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The main interface of REDAC is organized into three modules, each corresponding to a key step in the data analysis workflow:

1. **Perform a Complete Analysis:** executes the entire RNA-Seq analysis pipeline from start to finish.
2. **Enrichment Analysis and Result Interpretation:** focuses on the interpretation of enrichment results derived from differential expression analysis.
3. **Plot Generation:** allows users to generate individual plots independently of the full analysis workflow.

2 How to use the Perform a Complete Analysis module

The user interface (shown in Figure 1) constitutes the *Perform a Complete Analysis* module. This module guides the user through a complete bulk RNA-seq differential expression data analysis using raw count data as input via natural language interaction with the Gemma LLM.

To initiate the analysis workflow, users begin by clicking the “Browse” button to upload a tab-separated values (TSV) file containing the RNA-Seq count matrix (Figure 2). The input file format requires genes as rows and samples as columns, with raw count values representing expression levels.

Upon successful file upload, the data contents are immediately displayed for preliminary inspection. The interface provides flexible viewing options, allowing users to examine the first 10, 50, or 100 rows of the data set. A search functionality enables rapid gene-specific queries by entering gene identifiers in the designated search box (Figure 2).

After data upload, REDAC automatically generates a comprehensive suite of diagnostic visualizations (Figures 3 and 4) to facilitate quality assessment and identification of potential data anomalies, including sample mislabeling or aberrant count distributions. Users can navigate between different visualization types through the tabbed interface, as demonstrated in Figure 4 where the three-dimensional PCA plot is selected.

REDAC: RNA-seq Expression Data Analysis Chatbot

A Web App for analysing bulk RNA-seq data by asking questions written in English language

Perform a Complete Analysis

Enrichment Analysis and Result Interpretation

Plot Generation

You can use this chatbot by uploading your bulk RNA-seq raw count data (positive integers) in a tab separated format (as the one shown below).

Gene	ID27	ID33	ID68	ID70
SEC24B-AS1	47	3	0	26
A1BG0	410	3	14	4
A1CF	192	202	156	63
GGACT	28	23	17	15

... ..

Then, write a request and click on 'Run DE Analysis!' button below.

Please, upload your bulk RNA-seq raw count data (positive integers) in a tab separated format

Browse...

No file selected

Please, write your request (by Gemma)

Example: perform an rnaseq analysis between treated 3,4 and wt 1,2 samples, up regulated

Your Input

Show 10 entries

Search:

Showing 1 to 2 of 2 entries

Previous

1

Next

<< DATA INSPECTION PLOTS >>

Dendrogram

PCA Components

PCA

PCA 3D

Violin Plot

Densities

Heatmap

Figure 1: REDAC main interface.

The analysis is initiated through a natural language interface where users specify their experimental comparison of interest. This includes defining the sample groups for statistical testing using edgeR methodology and specifying the direction of differential expression analysis (up regulated, down regulated, or de regulated). For instance, a query might be: “execute an rnaseq analysis between treated 3,4 and wt 1,2 samples, de regulated” as the one shown in Figure 2 in red circle.

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Gene	ID27	ID33	ID68	ID70
SEC24B-AS1	47	3	0	26
A1BG0	410	3	14	4
A1CF	192	202	156	63
GGACT	28	23	17	15

Then, write a request and click on 'Run DE Analysis!' button below.

Please, upload your bulk RNA-seq raw count data (positive integers) in a tab separated format

Browse... expression_file.txt

Upload complete

Please, write your request (by Gemma)

execute an rnaseq analysis between treated 3,4 and wt 1,2 samples, de regulated

Your Input

Show **10** entries Search:

	ID27	ID33	ID68	ID70	ID21	ID44	ID64	ID19	ID20	ID30	ID39	ID60	ID4
A1BG	0	4103	14	4	10	13	5	7448	10033	0	177	4433	
A1CF	192	202	156	63	7	157	35	744	1474	150	104	1022	2
GGACT	28	23	17	26	36	4	5	24	14	10	17	10	
A2M	1300	7492	20369	8279	2479	586	7746	24483	33899	9158	2292	19986	16
A2ML1	0	7	934	0	3	1	0	0	0	0	0	0	
A2MP1	0	0	4	0	0	0	37	0	8	2	0	0	
A4GALT	173	11	1169	1016	101	18	294	66	45	298	194	148	1
A4GNT	0	1	0	0	3	0	0	1	4	0	1	3	
AAAS	1034	769	837	475	63	518	459	254	679	904	638	553	5
AACS	864	736	1297	639	397	171	699	116	382	681	951	1399	11

Showing 1 to 10 of 20,811 entries Previous **1** 2 3 4 5 ... 2,082 Next

Figure 2: File upload interface showing the Browse button (highlighted in red circle) for uploading tab-separated value files containing RNA-Seq count data. Users must specify their analytical request by defining sample comparisons for edgeR differential expression testing and indicating the direction of interest (up regulated, down regulated, or de regulated).



Figure 3: Automated quality assessment visualizations generated following gene expression file upload. These diagnostic plots enable users to identify potential data anomalies, including sample mislabeling or aberrant count distributions. The dendrogram shown here illustrates sample clustering based on expression profiles.

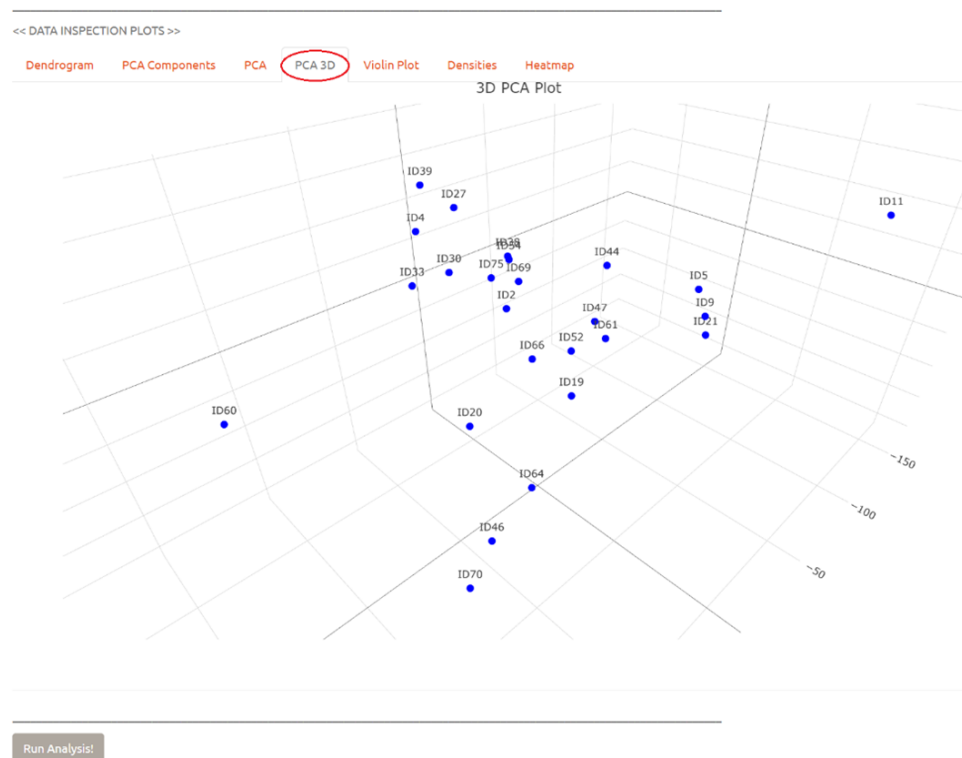


Figure 4: Interactive visualization interface demonstrating navigation between different plot types through tabbed selection. The three-dimensional principal component analysis (PCA 3D) tab is shown selected (highlighted in red circle).

Once the data have been inspected and the request has been written, we can click the Run

Analysis! button. Subsequently, a yellow pop-up appears on the screen with a waiting message and a progress bar is shown (see Figure 5).



Figure 5: Analysis execution interface showing the progress notification system activated upon selecting the Run Analysis! button. The yellow notification popup provides real-time status updates with an integrated progress bar.

When the execution is completed, the first 10 lines of results are shown. We can download the entire result table by clicking the Download the Result Table button (see Figure 6).

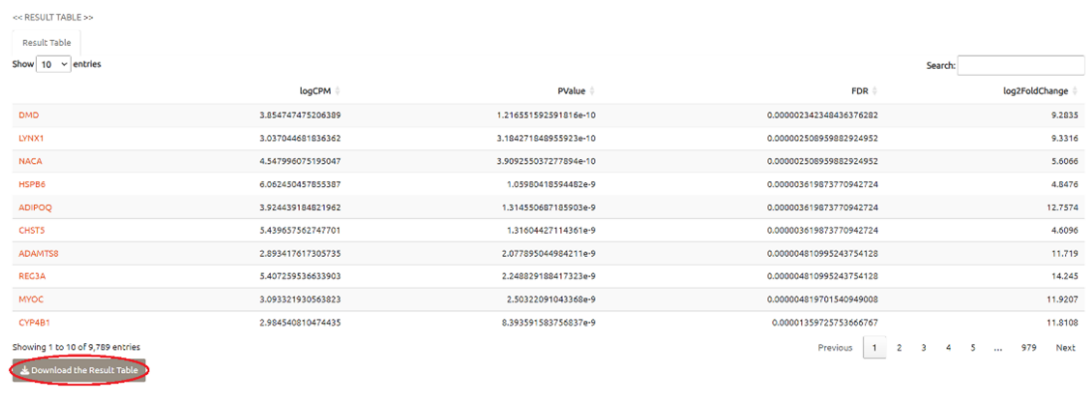


Figure 6: Results summary interface displaying the top 10 differentially expressed genes upon analysis completion. The complete results table can be downloaded using the designated button (highlighted in red circle).

Moreover, a Volcano Plot and a MA Plot are automatically generated to evaluate results globally (see Figure 7). If we put the mouse pointer on a red dot, the corresponding gene name will appear on the screen (see Figure 7).



Figure 7: Automated generation of volcano plot and MA plot visualizations for the evaluation of differential expression results. Interactive functionality allows gene identification through mouse-over display of gene names.

At this stage, REDAC provides multiple pathways for result interpretation and further analysis tailored to different user expertise levels. Users new to RNA-seq analysis can access the “Analysis Discussion (by Gemma)” feature to obtain a guided explanation of the results (see Figure 8). This LLM provides comprehensive descriptions of the typical analytical steps in RNA-seq workflows and recommends appropriate methodological approaches for each stage.

For experienced R users seeking greater analytical flexibility, the “Alternative code for R developers (by LLaMA)” option generates customized R scripts that implement the same analysis using alternative functions or parameter configurations (see Figure 9). This feature enables advanced users to explore different analytical approaches while maintaining reproducibility.

Additionally, users can generate a comprehensive, interactive HTML report by selecting “Download the HTML Report” (see Figure 7). This report provides complete analytical documentation including detailed descriptions of each analytical step, all generated visualizations, and the complete R code executed during the analysis. The HTML format ensures cross-platform compatibility and enables interactive exploration of results in any standard web browser (see Figures 10, 11, and 12).

 Download the Result Table

<< RESULT INSPECTION PLOTS, DISCUSSION AND ALTERNATIVE CODE>>

Volcano Plot

MA Plot

Analysis discussion (by Gemma)

Alternative code for R developers (by Llama)

Here's a general workflow for your RNA-seq analysis, incorporating the requested plots and tools:

1. Data Preparation and Quality Control

- **Load Data:** Import your data into R. Since you have gene IDs as row names, ensure your data is loaded correctly.
- **Create Metadata:** Create a data frame containing information about your samples (e.g., "treated" or "wt", sample IDs). This will be crucial for downstream analysis.
- **Exploratory Data Analysis:**
 - **PCA Plot:** Perform Principal Component Analysis (PCA) to visualize sample relationships. This can reveal potential batch effects or outliers.
 - **Density Plots:** Generate density plots for each sample to assess the distribution of read counts. Look for consistent shapes across samples.

2. Differential Gene Expression Analysis

- **Normalization:** Normalize your count data to account for library size differences between samples. Popular methods include:
 - **DESeq2:** A widely used package for RNA-seq analysis that offers robust normalization and differential expression testing.
 - **edgeR:** Another powerful package that uses a negative binomial distribution to model count data.
 - **limma:** Originally designed for microarray data, but adaptable to RNA-seq. It uses linear models for differential expression analysis.
- **Differential Expression Testing:** Use your chosen package (DESeq2, edgeR, or limma) to identify genes that are differentially expressed between your "treated" and "wt" groups.

3. Results Exploration and Visualization

- **MA Plot:** Create an MA plot to visualize the log2 fold change (M) against the average expression level (A) of genes. This helps identify genes with large fold changes.
- **Volcano Plot:** Generate a volcano plot to display the -log10(adjusted p-value) on the y-axis and the log2 fold change on the x-axis. This highlights statistically significant differentially expressed genes.
- **Heatmap:** Create a heatmap to visualize the expression patterns of your top differentially expressed genes across all samples. This can reveal clusters of co-regulated genes.
- **Dendrogram:** Construct a dendrogram based on gene expression distances to visualize the hierarchical relationships between samples.

4. Downstream Analysis

- **Gene Ontology (GO) Enrichment Analysis:** Use tools like GOSec or DAVID to identify enriched biological processes or pathways among your differentially expressed genes.
- **Pathway Analysis:** Explore pathway databases (e.g., KEGG, Reactome) to understand the functional implications of your findings.

Important Considerations:

- **Batch Effects:** Be aware of potential batch effects in your data and address them appropriately during normalization or analysis.
- **Multiple Testing Correction:** Apply multiple testing correction (e.g., Benjamini-Hochberg) to control the false discovery rate when identifying differentially expressed genes.
- **Biological Interpretation:** Always interpret your results in the context of your experimental design and biological question.

Let me know if you have any more specific questions about any of these steps!

 Download the HTML Report

 Download the Word Report

Figure 8: Analysis Discussion interface powered by Gemma LLM. This feature provides comprehensive explanations of RNA-seq analysis workflows, detailing the standard analytical steps and recommended methodological approaches for each stage.

<< RESULT INSPECTION PLOTS, DISCUSSION AND ALTERNATIVE CODE>>

Volcano Plot

MA Plot

Analysis discussion (by Gemma)

Alternative code for R developers (by Llama)

To perform an RNA-seq analysis between treated samples (3, 4) and wild-type (WT) samples (1, 2) using the DESeq2 package in R, we first need to ensure that our data is properly formatted and that we have the necessary libraries installed. The DESeq2 package is used for differential gene expression analysis based on the negative binomial distribution.

Given that you have a count matrix with 20,811 rows (genes) and 26 columns (samples), and you want to compare treated samples (let's say ID4, ID66, ID38, ID46, ID52, ID54, ID47, ID2, ID61, ID5, ID9, ID69, ID75, ID11 for treated, assuming the rest are WT or other conditions not of interest for this comparison) against WT samples (let's say ID27, ID33, ID68, ID70, ID21, ID44, ID64, ID19, ID20, ID30, ID39, ID60 for WT), we'll proceed with the analysis.

First, ensure you have DESeq2 installed. If not, you can install it using:

```
install.packages("DESeq2")
install.packages("BiocManager")
BiocManager::install("SummarizedExperiment")
```

Now, let's proceed with the analysis:

```
# Load necessary libraries
library(DESeq2)
library(data.table)

# Assuming 'countData' is your count matrix and 'sampleNames' are your sample names
# Create a sample table (you need to adjust this based on your actual sample information)
sampleTable <- data.frame(
  sampleName = c("ID27", "ID33", "ID68", "ID70", "ID21", "ID44", "ID64", "ID19", "ID20",
    "ID30", "ID39", "ID60",
    "ID4", "ID66", "ID38", "ID46", "ID52", "ID54", "ID47", "ID2", "ID61",
    "ID5", "ID9", "ID69", "ID75", "ID11"),
  condition = c(rep("WT", 12), rep("Treated", 14)) # Adjust based on actual conditions
)

# Create a DESeqDataSet object
dds <- DESeqDataSetFromMatrix(
  countData = countData,
  colData = sampleTable,
  design = ~ condition
)

# Perform differential expression analysis
dds <- DESeq(dds)

# Extract results
res <- results(dds)

# Filter for up-regulated genes with adjusted p-value < 0.1
upRegulated <- res[res$padj < 0.1 & res$log2FoldChange > 0, ]

# View up-regulated genes
head(upRegulated)
```

This code snippet assumes you have a count matrix named `countData` and that you've correctly identified which samples belong to the "WT" and "Treated" conditions. You'll need to adjust the `sampleTable` to accurately reflect the condition of each sample.

Also, note that the `design` formula in `DESeqDataSetFromMatrix` specifies the design of the experiment. In this case, we're comparing the "Treated" condition against the "WT" condition.

The `results` function is then used to extract the results of the differential expression analysis, and we filter these results to find genes that are up-regulated (i.e., have a positive log2 fold change and an adjusted p-value less than 0.1).

Remember, the choice of adjusted p-value threshold (here, 0.1) can be adjusted based on your specific research question and the stringency of filtering you desire.

[Download the HTML Report](#)
[Download the Word Report](#)

Figure 9: Alternative R code generation interface powered by LLaMA. This feature generates customized R scripts implementing equivalent analyses using alternative functions or parameter configurations, enabling experienced users to explore different analytical approaches.

Report of the Analysis

Design and Dispersion

```
Sys.time()
```

```
## [1] "2025-07-31 16:12:03 CEST"
```

```
counts <- params$counts
res <- params$res
#treated_samples <- colnames(counts)[as.numeric(params$treated_samples)]
#wt_samples <- colnames(counts)[as.numeric(params$wt_samples)]
#which_results <- params$which_results

# Build metadata
#treated_condition <- data.frame(patients = treated_samples, condition = 'treated')
#wt_condition <- data.frame(patients = wt_samples, condition = 'wt')
#samples <- rbind(wt_condition, treated_condition)
#rownames(samples) <- samples$patients

# Subset and order count matrix
#dataset <- counts[, samples$patients]
#samples$condition <- factor(samples$condition, levels = c('wt', 'treated'))

# Create DGEList
#dge <- DGEList(counts = dataset, group = samples$condition)

# Filter Low counts
#keep <- rowSums(counts) >= dim(counts)[2]
#dge <- dge[keep, , keep.lib.sizes = FALSE]

# Normalize and estimate dispersion
#dge <- calcNormFactors(dge)
#dge <- estimateDisp(dge)

# Differential expression
#et <- exactTest(dge, pair = c('wt', 'treated'))
#res <- topTags(et, n = nrow(dge))$table
#res <- res[order(res$FDR), ]
#res$log2FoldChange <- round(res$logFC, 4)
#res$logFC <- NULL

# Filter DE results
#if (which_results == 'down') {
#  Deres <- res[res$log2FoldChange < 0, ]
#} else if (which_results == 'up') {
#  Deres <- res[res$log2FoldChange > 0, ]
#} else {
#  Deres <- res
#}
#DERes <- na.omit(Deres)

# Plot
#plotBCV(dge)
```

Top 100 Differentially Expressed Genes

```
knitr::kable(head(res, 100), caption = 'Top 100 Differentially Expressed Genes')
```

Top 100 Differentially Expressed Genes

	logCPM	PValue	FDR	log2FoldChange
--	--------	--------	-----	----------------

Figure 10: Header section of the automatically generated HTML report, providing comprehensive analytical documentation.

```
#plotBCV(dge)
```

Top 100 Differentially Expressed Genes

```
knitr::kable(head(res, 100), caption = 'Top 100 Differentially Expressed Genes')
```

Top 100 Differentially Expressed Genes

	logCPM	PValue	FDR	log2FoldChange
DMD	3.854748	0.00e+00	0.0000023	9.2835
LYNX1	3.037045	0.00e+00	0.0000025	9.3316
NACA	4.547996	0.00e+00	0.0000025	5.6066
HSPB6	6.062450	0.00e+00	0.0000036	4.8476
ADIPOQ	3.924439	0.00e+00	0.0000036	12.7574
CHST5	5.439658	0.00e+00	0.0000036	4.6096
ADAMTS8	2.893418	0.00e+00	0.0000048	11.7190
REG3A	5.407260	0.00e+00	0.0000048	14.2450

Figure 11: Top 100 differentially expressed genes as displayed in the interactive HTML report, showing detailed statistical results and gene annotations.

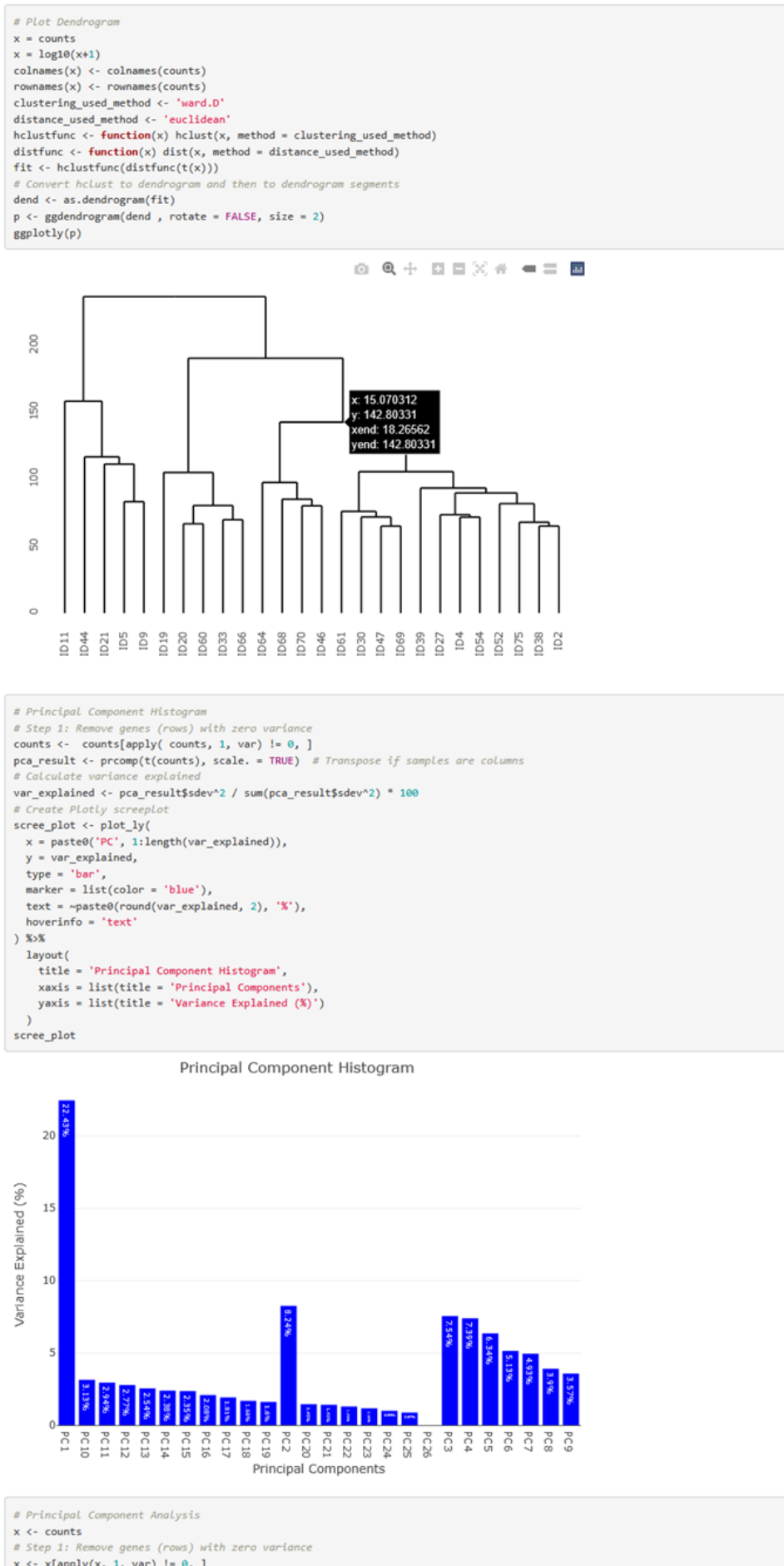


Figure 12: Representative section of the HTML report demonstrating the analytical documentation format.

Upon downloading the differential expression results table, users can navigate to the second tab, “Enrichment Analysis and Result Interpretation”, to upload this file for subsequent pathway enrichment analysis.

3 How to use the Enrichment Analysis and Result Interpretation module

This module facilitates biological interpretation of differential expression results through pathway enrichment analysis. Users initiate the process by clicking the “Browse” button to upload an edgeR result file, as demonstrated in Figure 13. Following successful file upload, the interface displays the first 10 lines of the input data for verification purposes.

To enhance the contextual accuracy of the LLM-generated interpretations, users should provide a descriptive request that specifies the experimental context and biological question. This contextual information enables the LLMs to generate more precise and relevant interpretations of the enrichment results. For example, a comprehensive request might state: *“Explain these results produced from the comparison of pulse-induced drug resistance in HCC827 cells against their parental cells to study the mechanisms induced by this suboptimal dosage of Gefitinib”* (Figure 13). After entering the contextual description, users initiate the analysis by clicking the “Enrich!” button.

The system then performs KEGG pathway enrichment analysis on the uploaded differential expression data. Upon completion, the enrichment results are displayed in a tabular format (Figure 14). Users can visualize the most significantly enriched KEGG pathways by selecting the “Dot Plot” tab (Figure 15), which generates a comprehensive dot plot visualization displaying pathway enrichment statistics.

REDAC: RNA-seq Expression Data Analysis Chatbot

A Web App for analysing bulk RNA-seq data by asking questions written in English language

Perform a Complete Analysis Enrichment Analysis and Result Interpretation Plot Generation

You can perform an enrichment analysis by uploading your edgeR result file in a tab separated format (as the one shown below).

	logCPM	PValue	FDR	log2FoldChange
SEC24B-AS1	3.86	2.1e-12	...	...
A1BG0	3.04	1.3e-11	...	...
A1CF	6.07	1.1e-10	...	...

Finally, this chatbot can suggest an interpretation of your results via two LLMs, such as: Gemma and Llama.

Please, upload your edgeR result file in a tab separated format

Browse... result_of_perform_rnaseq_analysis_between_treated_7,8_and_wt_1,2_A_samples_regulated_up_on_Raw_data.txt

Upload complete

Write your request for the LLMs:

Explain these results produced from the comparison of pulse-induced drug resistance in HCC827 cells against their parental cells to study the mechanisms induced by this suboptimal dosage of Gefitinib

Enrich!

Data Tables

Your Input Enrichment Results Dot Plot A possible interpretation (Gemma): Another possible interpretation (Llama):

Show 10 entries

	logCPM	PValue	FDR	log2FoldChange
S100A9	9.09214109463893	0	0	10.8496
SERPINA3	8.01417996761387	0	0	9.9682
C3	6.74576724924454	0	0	9.0585
CCL20	6.39588961433587	0	0	8.4443
S100P	8.43815118078001	0	0	8.4093
C1S	6.81667604204262	0	0	8.3743
CP	7.47188501781367	0	0	7.8101
CHI3L2	6.21708615147535	0	0	6.9009
CXCL8	8.63340312270837	0	0	6.4801
SAA1	7.4198054986928	0	0	6.2936

Showing 1 to 10 of 9,498 entries

Download Enrichment Results

Previous 1 2 3 4 5 ... 950 Next

Figure 13: The enrichment analysis module interface showing the file upload functionality for edgeR differential expression results. Users upload their results file via the Browse button, with the first 10 lines displayed for verification. A contextual prompt is provided to guide LLM interpretation of the enrichment analysis. The example prompt reads: “Explain these results produced from the comparison of pulse-induced drug resistance in HCC827 cells against their parental cells to study the mechanisms induced by this suboptimal dosage of Gefitinib”. Analysis is initiated using the Enrich! button (highlighted in red).

Please, upload your edgeR result file in a tab separated format

Browse... result_of_perform rnaseq analysis between treated 7,8 and wt 1,2 A samples, regulated_up_on_Raw_data.txt

Upload complete

Write your request for the LLMs:

Explain these results produced from the comparison of pulse-induced drug resistance in HCC827 cells against their parental cells to study the mechanisms induced by this suboptimal dosage of Gefitinib

Enrich!

Data Tables

Your Input **Enrichment Results** Dot Plot A possible interpretation (Gemma): Another possible interpretation (Llama):

Show 10 entries Search:

	Description	subcategory	GeneRatio	p.adjust	qvalue	geneID
hsa04668	TNF signaling pathway	Signal transduction	68/2118	6.771553618751005e-14	3.800204518334403e-14	6364/6372/6374/2919/3553/2921/3383/5743/7133/330/1051/9021/2920/4792/843/79651/60
hsa04010	MAPK signaling pathway	Signal transduction	122/2118	8.993250686137448e-11	5.047023743168294e-11	3552/3553/10125/3643/1942/7422/3815/1848/8912/4791/1847/59285/2261/2260/4217/1321
hsa04657	IL-17 signaling pathway	Immune system	53/2118	1.791422023685258e-10	1.005348322104574e-10	6364/3576/6372/6374/3934/2919/3553/2921/5743/1051/6279/1673/4586/6278/2920/4792/7
hsa05165	Human papillomavirus infection	Infectious disease: viral	127/2118	2.996964598035465e-9	1.681900350785867e-9	5743/5522/3678/4600/4599/9636/3455/1298/7422/8324/3914/3696/64764/9641/10312/677:
hsa04148	Efferocytosis	Transport and catabolism	72/2118	3.728594557815463e-9	2.092492016368453e-9	5743/1051/5031/6446/834/1847/4035/140885/961/100133941/4776/1846/23250/5175/5777
hsa05168	Herpes simplex virus 1 infection	Infectious disease: viral	78/2118	2.80167869898129e-8	1.572305655427085e-8	718/3553/4939/330/972/4938/3678/9021/3455/4792/4940/684/3569/9641/567/6773/637/67
hsa04151	PI3K-Akt signaling pathway	Signal transduction	131/2118	4.927451862946963e-8	2.76529226344665e-8	5522/6446/118788/3678/3643/3455/1942/1298/7422/3815/54541/3914/3569/9180/2261/36
hsa04621	NOD-like receptor signaling pathway	Immune system	79/2118	6.70842292322266e-8	3.764775491570429e-8	3576/2919/3553/2921/4939/330/1673/2634/4938/23710/5330/10628/2920/834/3455/2633/2
hsa04015	Rap1 signaling pathway	Signal transduction	86/2118	6.70842292322266e-8	3.764775491570429e-8	3643/5330/5900/1942/7422/3815/5909/6494/2261/112/2260/2770/9771/5606/10411/6714/2
hsa04510	Focal adhesion	Cellular community - eukaryotes	83/2118	7.439252736312915e-8	4.174918113811283e-8	330/3678/1298/7422/3914/3696/6696/56924/3694/2335/6714/2889/3918/3915/1282/7410/3

Showing 1 to 10 of 91 entries

Download Enrichment Results

Previous 1 2 3 4 5 ... 10 Next

Figure 14: KEGG pathway enrichment analysis results interface. Upon completion of the analysis, users can access the enrichment results through the Enrichment Results tab (highlighted in red circle) to view the statistical output, and utilize the Download Enrichment Results button (highlighted in red circle) to export the complete results table for further analysis.

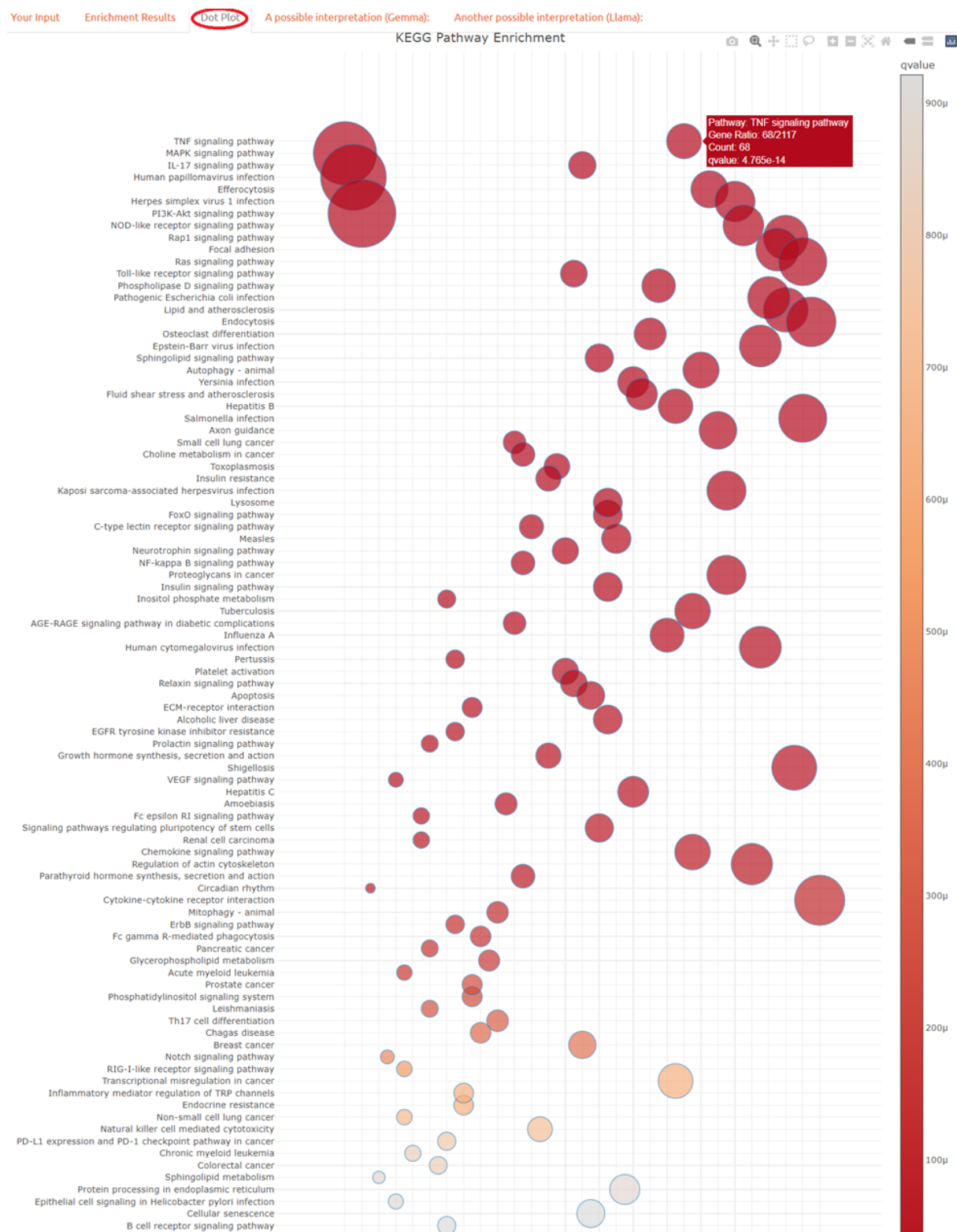


Figure 15: Selection of the Dot Plot tab (highlighted in red circle) generates a comprehensive dot plot visualization of the enriched pathways.

For biological interpretation of the enriched KEGG pathways, REDAC provides AI-powered analysis through two complementary large language models. Users can access the primary interpretation by selecting the “*A possible interpretation (Gemma)*” tab (Figure 16). This feature

leverages the Gemma model to generate contextual biological insights based on the enrichment results and the experimental context provided during the initial analysis setup. For comparative analysis or alternative perspectives, users can obtain a secondary interpretation by selecting the “Another possible interpretation (LLaMA)” tab (Figure 17).

Write your request for the LLMs:

Explain these results produced from the comparison of pulse-induced drug resistance in HCC827 cells against

Enrich!

Data Tables

Your Input Enrichment Results Dot Plot **A possible interpretation (Gemma)** Another possible interpretation (LLaMA)

Interpreting Pathways Involved in Gefitinib Pulse-Induced Drug Resistance

The list you provided represents a diverse array of pathways potentially implicated in Gefitinib pulse-induced drug resistance in HCC827 cells compared to their parental counterparts. Let's categorize these pathways and propose some plausible cellular mechanisms:

I. Signaling Pathways and Drug Resistance:

- **MAPK, PI3K-Akt, Ras, Rap1, EGFR, ErbB, Notch, NF-kappa B, FoxO, insulin, Phospholipase D, Sphingolipid:** These pathways are well-known players in cell survival, proliferation, and response to stress.
 - **Possible Mechanisms:** Pulse-induced Gefitinib exposure might select for cells with constitutive activation of these pathways, bypassing the drug's inhibitory effect on EGFR. This could involve mutations, gene amplification, or altered expression of pathway components.
- **Cytokine-cytokine receptor interaction, Chemokine signaling, TNF, IL-17:** These pathways are involved in inflammation and immune response.
 - **Possible Mechanisms:** Gefitinib pulse might trigger an inflammatory response, leading to the release of cytokines and chemokines that promote cell survival and proliferation, contributing to drug resistance.

II. Immune Evasion and Viral Infection:

- **Human papillomavirus infection, Herpes simplex virus 1 infection, Epstein-Barr virus infection, Kaposi sarcoma-associated herpesvirus infection, Human cytomegalovirus infection, Measles, Influenza A, Pertussis, Toxoplasmosis, Leishmaniasis, Chagas disease:**
 - **Possible Mechanisms:** These viral infections often manipulate host cell signaling pathways to evade immune surveillance and promote their own survival. Gefitinib pulse might inadvertently create a more permissive environment for these viruses, leading to increased viral load and potentially contributing to drug resistance.

III. Cellular Processes and Stress Response:

- **Autophagy, Mitophagy, Endocytosis, Efferocytosis, Proteoglycans in cancer, Protein processing in endoplasmic reticulum:**
 - **Possible Mechanisms:** Gefitinib pulse might induce cellular stress, triggering autophagy and mitophagy as survival mechanisms. Altered endocytosis and efferocytosis could also play a role in removing damaged cellular components and promoting cell survival.
- **Apoptosis, Cellular senescence:**
 - **Possible Mechanisms:** Gefitinib pulse might initially induce apoptosis in sensitive cells, but resistant cells might develop mechanisms to evade apoptosis or enter a state of senescence, allowing them to survive.

IV. Metabolic Reprogramming:

- **Choline metabolism in cancer, Glycerophospholipid metabolism, Sphingolipid metabolism, Lipid and atherosclerosis:**
 - **Possible Mechanisms:** Gefitinib pulse might alter cellular metabolism, favoring pathways that support cell survival and proliferation under stress. This could involve changes in lipid metabolism, choline utilization, and sphingolipid signaling.

V. Other Pathways:

- **Focal adhesion, Axon guidance, ECM-receptor interaction, Regulation of actin cytoskeleton:**
 - **Possible Mechanisms:** These pathways are involved in cell adhesion, migration, and cytoskeletal organization. Gefitinib pulse might induce changes in these pathways, allowing resistant cells to adapt to the drug-induced stress and potentially metastasize.
- **Transcriptional misregulation in cancer, Endocrine resistance, Inflammatory mediator regulation of TRP channels:**
 - **Possible Mechanisms:** Gefitinib pulse might lead to epigenetic changes or mutations that alter gene expression, contributing to drug resistance. Endocrine resistance mechanisms might also be activated, allowing cells to bypass the drug's effects.

Future Investigations:

1. **Validate Key Pathways:**
 - Use targeted inhibitors or genetic manipulation (e.g., siRNA, CRISPR) to confirm the role of specific pathways in Gefitinib resistance.
2. **Identify Molecular Drivers:**
 - Perform RNA sequencing and proteomics to identify differentially expressed genes and proteins in resistant cells compared to parental cells.
3. **Investigate Epigenetic Changes:**
 - Analyze DNA methylation and histone modifications to determine if epigenetic alterations contribute to drug resistance.
4. **Explore the Role of the Microenvironment:**
 - Investigate how the tumor microenvironment (e.g., stromal cells, immune cells) influences Gefitinib resistance.
5. **Develop Novel Therapeutic Strategies:**
 - Based on the identified mechanisms, develop new therapeutic strategies to overcome Gefitinib resistance.

By systematically investigating these pathways and their underlying molecular mechanisms, we can gain a deeper understanding of Gefitinib resistance and develop more effective treatment strategies for lung cancer.

Figure 16: The Gemma-powered interpretation interface accessible through the “A possible interpretation (Gemma)” tab provides contextual biological analysis of the enriched KEGG pathways.

Write your request for the LLMs:

Explain these results produced from the comparison of pulse-induced drug resistance in HCC827 cells against :

Enrich!

Data Tables

Your Input Enrichment Results Dot Plot A possible interpretation (Gemma): **Another possible interpretation (LLaMA):**

The comparison of pulse-induced drug resistance in HCC827 cells against their parental cells has revealed a complex interplay of various signaling pathways. To provide a comprehensive understanding, I have grouped these pathways into categories and offered plausible cellular mechanistic explanations.

Category 1: Cell Signaling and Survival Pathways

- PI3K-Akt signaling pathway: Activation of this pathway may contribute to cell survival and resistance to Gefitinib by promoting anti-apoptotic signals.
- MAPK signaling pathway: This pathway may be involved in the regulation of cell proliferation, differentiation, and survival, potentially contributing to drug resistance.
- NF-kappa B signaling pathway: Activation of NF-kappa B may promote cell survival and resistance to apoptosis, leading to the development of drug resistance.
- Ras signaling pathway: Mutations or alterations in the Ras pathway may contribute to the development of drug resistance by promoting cell proliferation and survival.

Category 2: Inflammation and Immune Response

- TNF signaling pathway: Activation of TNF signaling may contribute to the development of drug resistance by promoting inflammation and immune evasion.
- IL-17 signaling pathway: This pathway may be involved in the regulation of inflammation and immune response, potentially contributing to drug resistance.
- Toll-like receptor signaling pathway: Activation of TLR signaling may promote inflammation and immune response, leading to the development of drug resistance.
- NOD-like receptor signaling pathway: This pathway may be involved in the regulation of inflammation and immune response, potentially contributing to drug resistance.

Category 3: Cell Cycle and Apoptosis

- Apoptosis: Dysregulation of apoptosis may contribute to the development of drug resistance by allowing cancer cells to evade programmed cell death.
- Autophagy - animal: Autophagy may play a role in the development of drug resistance by promoting cell survival and recycling of cellular components.
- Mitophagy - animal: Mitophagy may be involved in the regulation of cellular metabolism and energy production, potentially contributing to drug resistance.

Category 4: Viral Infections and Cancer

- Human papillomavirus infection: HPV infection may contribute to the development of cancer and drug resistance by promoting cell proliferation and survival.
- Epstein-Barr virus infection: EBV infection may be involved in the regulation of cell proliferation and survival, potentially contributing to drug resistance.
- Hepatitis B: HBV infection may contribute to the development of liver cancer and drug resistance by promoting cell proliferation and survival.

Category 5: Metabolism and Lipid Signaling

- Lipid and atherosclerosis: Alterations in lipid metabolism may contribute to the development of drug resistance by promoting cell survival and proliferation.
- Sphingolipid signaling pathway: This pathway may be involved in the regulation of cell survival and apoptosis, potentially contributing to drug resistance.
- Choline metabolism in cancer: Alterations in choline metabolism may contribute to the development of cancer and drug resistance by promoting cell proliferation and survival.

Category 6: Cancer-Specific Pathways

- EGFR tyrosine kinase inhibitor resistance: Activation of alternative signaling pathways may contribute to the development of resistance to EGFR inhibitors.
- Non-small cell lung cancer: Alterations in specific signaling pathways may contribute to the development of NSCLC and drug resistance.
- Breast cancer: Alterations in specific signaling pathways may contribute to the development of breast cancer and drug resistance.

Future investigations should focus on the following areas:

- Elucidating the molecular mechanisms:** Further studies are needed to elucidate the molecular mechanisms underlying the development of drug resistance in HCC827 cells.
- Identifying key drivers of resistance:** Identifying the key drivers of resistance will help in the development of targeted therapies to overcome resistance.
- Investigating the role of non-coding RNAs:** Non-coding RNAs, such as microRNAs and lncRNAs, may play a crucial role in the development of drug resistance.
- Developing combination therapies:** Developing combination therapies that target multiple signaling pathways may help to overcome drug resistance.
- Investigating the role of the tumor microenvironment:** The tumor microenvironment may play a crucial role in the development of drug resistance, and further studies are needed to investigate this area.
- Developing biomarkers for resistance:** Developing biomarkers for resistance will help in the early detection of resistance and the development of personalized therapies.
- Investigating the role of epigenetic modifications:** Epigenetic modifications, such as DNA methylation and histone modification, may play a crucial role in the development of drug resistance.
- Developing therapies that target cancer stem cells:** Cancer stem cells may be responsible for the development of drug resistance, and further studies are needed to investigate this area.

Download Enrichment Results

Figure 17: Alternative pathway interpretation using LLaMA can be accessed through the “Another possible interpretation (LLaMA)” tab (highlighted in red circle), enabling comparative analysis of enrichment results.

4 How to use the Plot Generation module

In this module enables users to create individual visualizations independently of the complete analysis workflow. This module supports two distinct input types: raw count matrices and differential expression results from edgeR analysis.

For count matrix input files, the following visualization types are available:

- boxplot,

- violin plot,
- heatmap,
- correlation heatmap,
- principal component analysis (PCA),
- three-dimensional PCA (3D PCA),
- hierarchical clustering dendrogram,
- density plot,
- PCA component loadings,
- gene interaction network,
- surface plot.

For edgeR differential expression result files, the module provides:

- volcano plot,
- MA plot,
- dot plot,
- KEGG pathway network (KEGGnet).

To demonstrate the network visualization tool, Figure 18 presents a gene interaction network generated from expression data in the input file `Raw_data.txt`. The analysis was initiated by entering “create a network” in the request field, which automatically triggered integration with the STRING database (<https://string-db.org/>). The STRING database provides seven categories of interaction evidence: text mining, experimental validation, curated databases, co-expression analysis, chromosomal neighborhood, gene fusion events, and phylogenetic co-occurrence.

Our implementation focuses exclusively on experimentally validated interactions, as these represent high-confidence physical protein-protein interactions suitable for integration with transcriptomic data. The STRING experimental interaction data are pre-integrated within REDAC, eliminating the need for manual data upload. To ensure analytical rigor, we applied a stringent confidence threshold, selecting only interactions with scores exceeding 980 (representing high-confidence associations). For each gene pair connected through such validated experimental evidence, we computed Pearson correlation coefficients using their respective expression profiles. Gene pairs exhibiting zero variance, which would result in undefined correlation values, were systematically excluded from the analysis.

The resulting network visualization employs a color-coded edge system where red edges denote positive correlations and blue edges indicate negative correlations. Edge color intensity corresponds to correlation strength: deep red indicates strong positive correlation, while intense blue represents strong negative correlation.

This interactive network visualization facilitates detailed exploration of gene co-expression patterns, enabling users to identify clusters of coordinately expressed genes and anti-correlated gene pairs that exhibit inverse expression relationships. Such network-based approaches can reveal regulatory relationships and functional modules that may not be apparent through conventional

pathway enrichment analyses, thereby providing complementary insights into the underlying biological systems by leveraging experimentally validated protein interaction data from the STRING database.

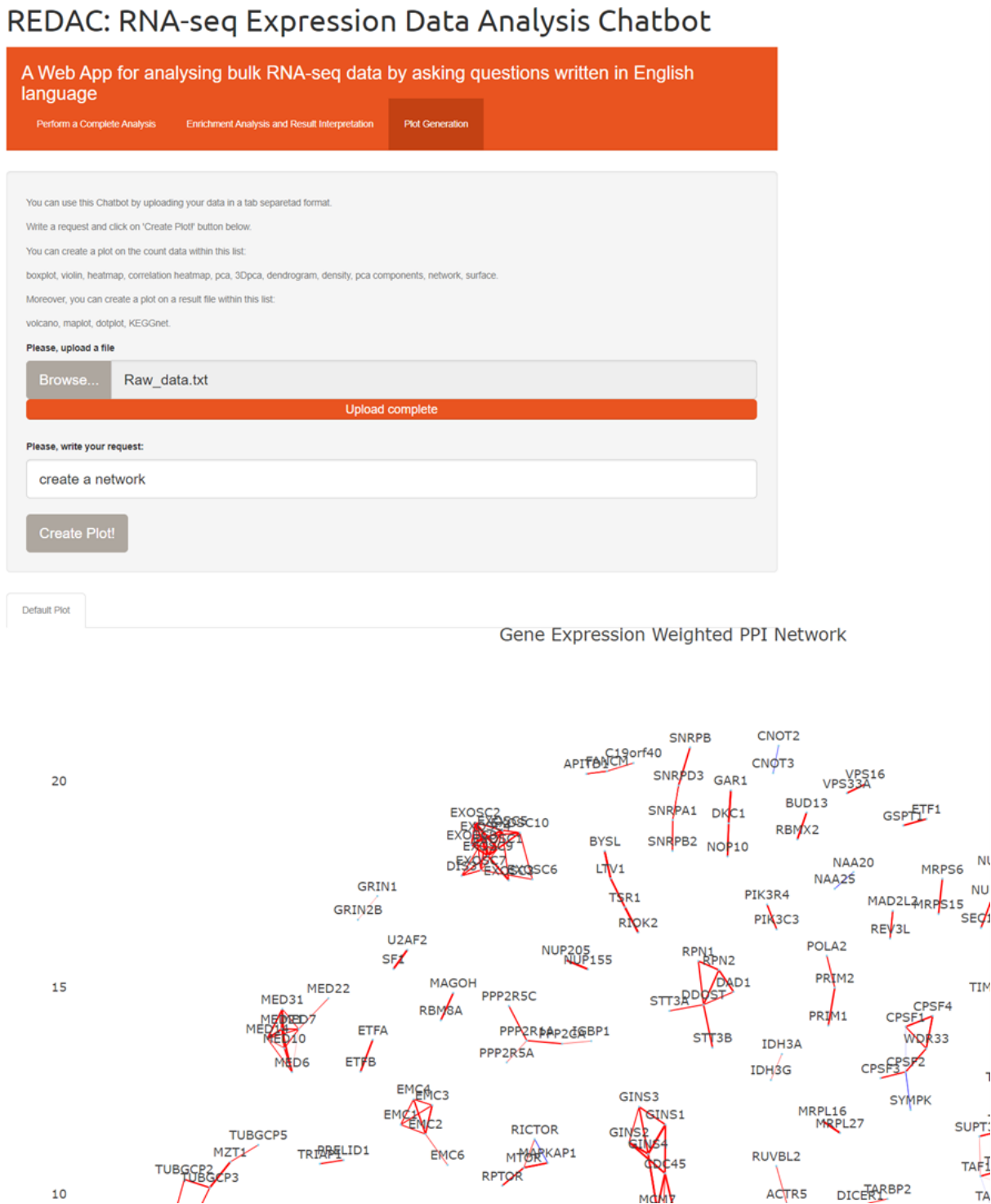


Figure 18: Gene interaction network integrating expression data with experimentally validated protein-protein interactions from the STRING database. Edge colors indicate expression correlations: red (positive), blue (negative), with intensity reflecting correlation strength.